XRPL Validator — Daily Engineering Report · 2026-08-09

Period: 2026-08-09 00:00 to 24:00 US Eastern (2026-08-09 04:00 to 2026-08-10 04:00 UTC). Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: xrpl-validator merge records (they begin 2026-07-04). Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Step 1 receipt records (they begin 2026-08-29); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); amendment data; data.xrpl.org agreement scores (scored days begin 2026-04-04; some days have none).

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** no receipt records for this day (they begin 2026-08-29).
- **Changes:** 0 deployments, 0 commits merged into main.

Service levels

No signing records for this day (they begin 2026-09-19).

Native engine conformance

No receipt records for this day (Step 1 records begin 2026-08-29, Steps 2–5 records 2026-10-09).

Live counters are kept only for the current run; none exist for a past day.

Receipt register

No receipt records for this day (Step 1 records begin 2026-08-29).

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

No deployment records before 2026-09-11.

Merged into main

No merges into main.

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **fix(xrpl-ledger): rank payment strands by UPPER BOUND, not by a realised trial** (`0f55ba5`, xrpl-validator) `payment.rs`'s round loop trialled every strand, measured what each pass actually delivered, and picked the best spent-per-delivered. That is the model this repo already REFUTED for offer crossing in 03c2cb9 -- `ActiveStrands::activateNext` sorts by `qualityUpperBound` best-first, DROPS any strand whose bound misses limitQuality, and the loop takes the first candidate whose pass is not rejected, `continue`-ing past …
- **feat(xrpl-node): FFI_VALCHECK — the oracle rule now works at VALUE level** (`c7d4625`, xrpl-validator) `PROBE: CLEAN` has only ever meant libxrpl touched the same OBJECTS mainnet did. It never said libxrpl wrote the same NUMBERS, because the entire parity verdict is a mutation-KEY comparison. So when 9bc95c8 made value divergences visible on the native side, there was no way to attribute them: 175 value hits and no verdict on whether any of them were ours.
- **fix(xrpl-ledger): token escrow — Cancel and Finish never released the token** (`6ef30c4`, xrpl-validator) #106179351: 413 txs, 221 matched. `EscrowCancel attempted=192 MATCH=0 DIVERGE-MUT=192` -- every EscrowCancel in the ledger, all the same shape, ours 3 mutations against mainnet's 8. FFI PROBE: CLEAN, so ours.
- **fix(xrpl-node): prefetch NFT pages for NFTokenModify — it is tt 61, not 25..=29** (`5aad67e`, xrpl-validator) `is_nft_tx_type` was `(25..=29).contains(&tt)` — Mint, Burn, CreateOffer, CancelOffer, AcceptOffer. NFTokenModify (XLS-46) is tt 61: it POSTDATES that contiguous range, so a transaction whose entire job is to locate a token and rewrite its URI never had the owner's NFTokenPages fetched. libxrpl's `nft::findToken` then searched a page set that was never loaded and returned tecNO_ENTRY on a token that plainly exists.
- **feat(xrpl-ledger): DX_CLAMP — and it retires the ceil_in flip for good** (`9e475fe`, xrpl-validator) The 2026-08-08 attempt to flip the taker-clamped XRP floor to a ceil regressed #105284280 (78/78 -> 77/78) and took #105091578 from 0 value hits to 21. The note left behind said the next attempt had to establish, first, whether the twelve one-drop-short residuals reach that branch at all.

- **fix(xrpl-ledger): the bridged XRP mid-leg must CEIL to whole drops, not truncate** (`9bb5ceb`, xrpl-validator) Autobridged crossing routes value through XRP, and that mid-leg has to land on whole drops. `cross_bridged` truncated it — `me_rescale(xrp, 0, false)` — so one leg moved a drop more than the other could absorb and the pass came out skewed in both directions at once.

Independence program

No independence record for UTC day 2026-08-09; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

No amendment records for this day.

Open items

None.

Corrections to earlier reports

None.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Stealth Meta-Address Fields on AccountRoot** (privacy) Add two optional SFields to AccountRoot — sfStealthScanPubKey and sfStealthSpendPubKey (compressed secp256k1 points) — settable via AccountSet after a
- **Native Subscription Pull-Payment Ledger Object** (payments) Add a new owner-reserved ledger object ltSUBSCRIPTION (keylet::subscription(payer, sequence)) holding sfPayee, sfMaxAmount, sfPeriod (ledgers between
- **Weighted Multi-Asset AMM Basket Pools** (amm-defi) Extend AMMCreate with an optional STArray sfAssets (2–8 entries, each an STObject of sfAsset + sfWeight in basis points summing to 10000) gated by a n

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
08-09 17:22	—	xrpl-validator	`0f55ba5`	fix(xrpl-ledger): rank payment strands by UPPER BOUND, not by a realised trial
08-09 19:17	—	xrpl-validator	`c7d4625`	feat(xrpl-node): FFI_VALCHECK — the oracle rule now works at VALUE level
08-10 01:59	—	xrpl-validator	`6ef30c4`	fix(xrpl-ledger): token escrow — Cancel and Finish never released the token
08-10 02:47	—	xrpl-validator	`5aad67e`	fix(xrpl-node): prefetch NFT pages for NFTokenModify — it is tt 61, not 25..=29
08-10 02:50	—	xrpl-validator	`9e475fe`	feat(xrpl-ledger): DX_CLAMP — and it retires the ceil_in flip for good
08-10 03:53	—	xrpl-validator	`9bb5ceb`	fix(xrpl-ledger): the bridged XRP mid-leg must CEIL to whole drops, not truncate

Times in UTC. A commit appears on the day it was committed to main or merged into it.